Generalized rootfinding: historical checkpoint

Historical Python provenance; source and input identities are pinned by TRANSPORT.json and the two frozen SOURCE_MANIFEST.json files. This PR introduces no new solver measurements.

Host: managed Linux x86_64 cloud worker, one CPU and 4 GiB worker budget. Qualified uncontended isolation, A/A control and independent external review were not established. Values are descriptive; no confidence interval was recorded.

All three frozen workers passed. Source snapshots and raw stdout/stderr/process/preregistration/result records are preserved byte-for-byte after gzip decoding. The full older journal is identified by digest but not bundled; this is not a complete resumable lab history.

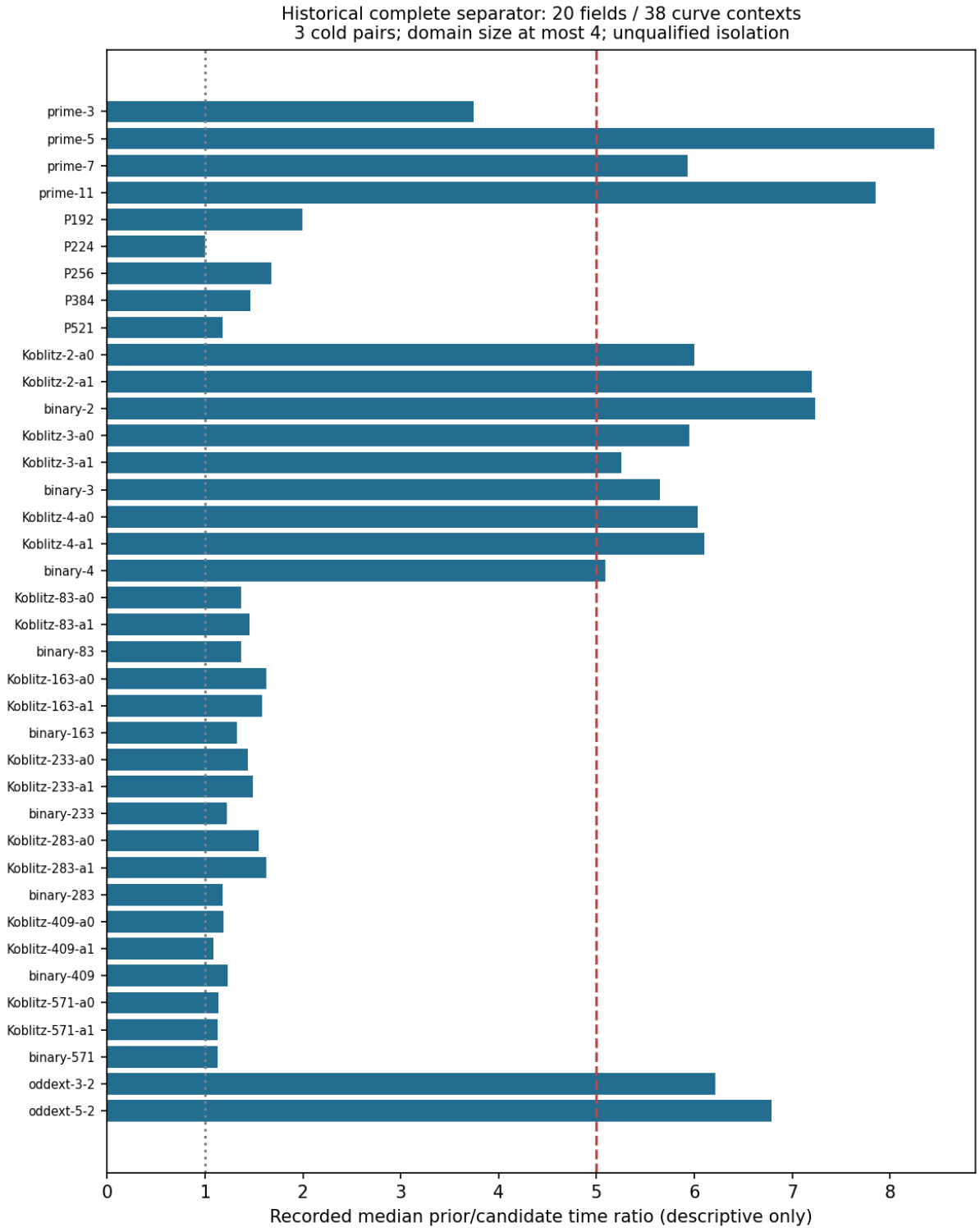
Requested target remains 5x against genuine matched all-roots F4/F5/F6, with the original immutable release gate. That goal is incomplete.

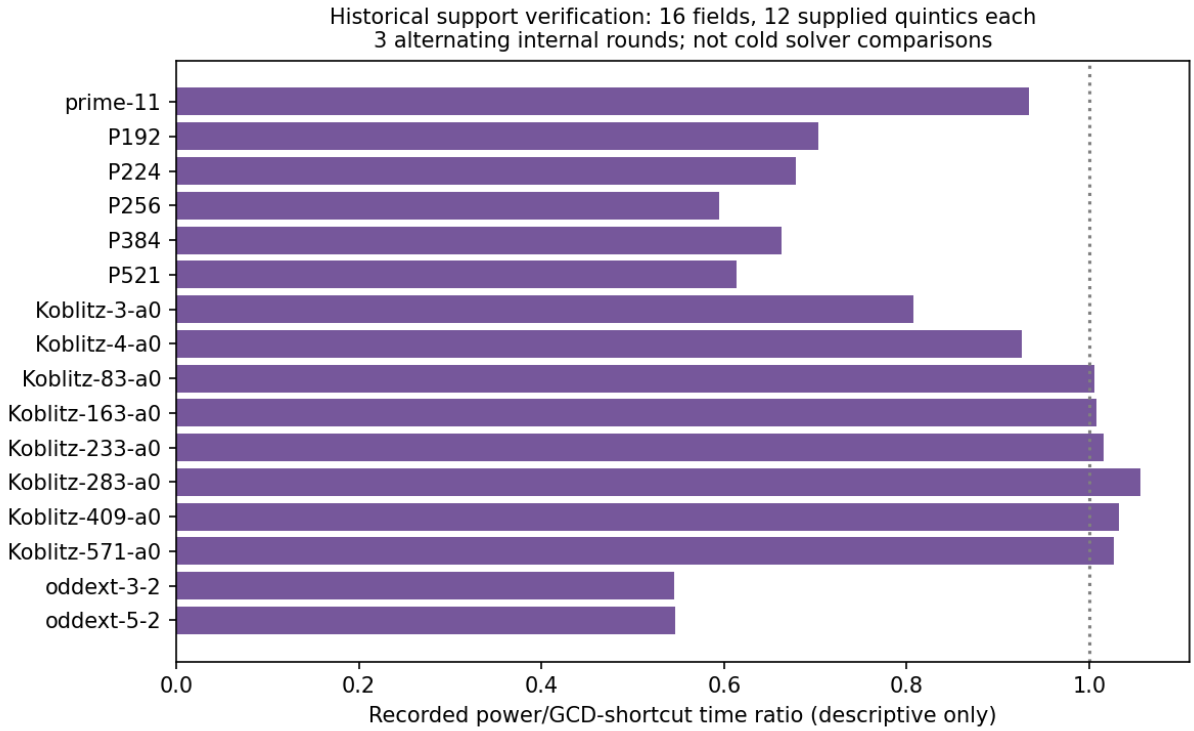
Batch calibration: whole-cold median ratio 1.3830038110595322 against the prior cached separator, 20 fields / 38 curve contexts, three cold pairs. Outputs agree across arms. Tiny domains do not establish hardness.

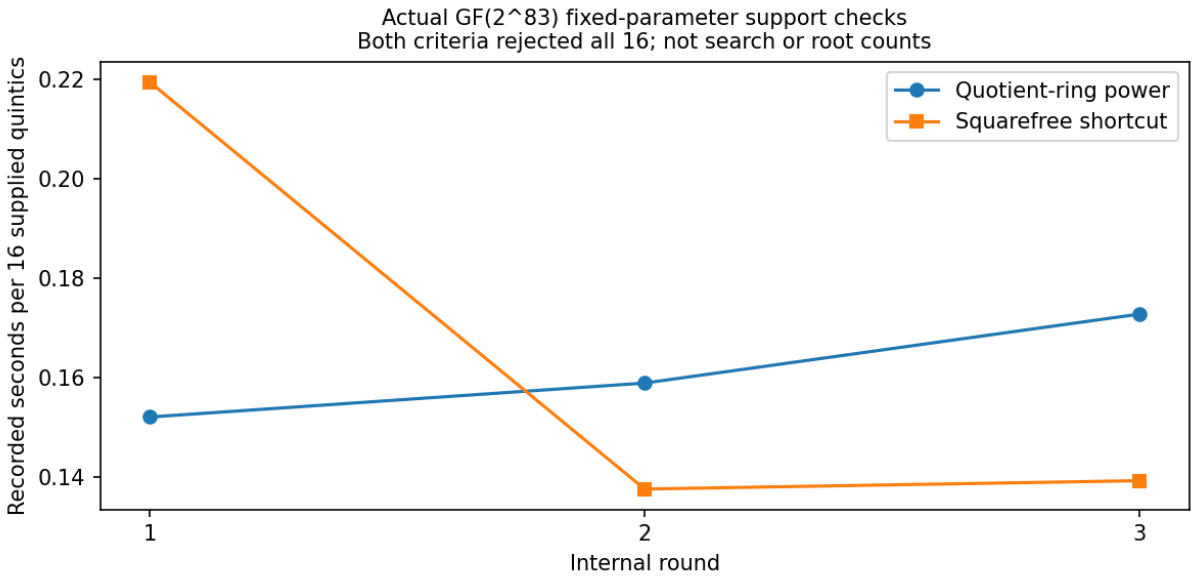
Support checks: actual GF83 fixed-parameter 16-quintic ratio 1.1408942779538267; both criteria rejected all candidates. Several NIST prime and odd-extension contexts regress under the blanket squarefree shortcut. No original-root count or search completion is claimed.

Historical independent oracle algorithms checked 15 complete small systems and 4,448 support cases. This native archival audit checks payload hashes, pinned closure and recorded paired output equality; it does not independently re-solve equations.

Next work: native solver/harness port before new research comparisons; separately test inversion-free squarefree detection and bounded sparse contraction width for the hard support search. No novel-priority, asymptotic, security or IC/rho claim.


Historical complete separator: 20 fields / 38 curve contexts
3 cold pairs; domain size at most 4; unqualified isolation
prime-3
prime-5
prime-7
prime-11
P192
P224
P256
P384
P521
Koblitz-2-a0
Koblitz-2-a1
binary-2
Koblitz-3-a0
Koblitz-3-a1
binary-3
Koblitz-4-a0
Koblitz-4-a1
binary-4
Koblitz-83-a0
Koblitz-83-a1
binary-83
Koblitz-163-a0
Koblitz-163-a1
binary-163
Koblitz-233-a0
Koblitz-233-a1
binary-233
Koblitz-283-a0
Koblitz-283-a1
binary-283
Koblitz-409-a0
Koblitz-409-a1
binary-409
Koblitz-571-a0
Koblitz-571-a1
binary-571
oddext-3-2
oddext-5-2
0
1
2
3
4
5
6
7
8
Recorded median prior/candidate time ratio (descriptive only)


Historical support verification: 16 fields, 12 supplied quintics each
3 alternating internal rounds; not cold solver comparisons
prime-11
P192
P224
P256
P384
P521
Koblitz-3-a0
Koblitz-4-a0
Koblitz-83-a0
Koblitz-163-a0
Koblitz-233-a0
Koblitz-283-a0
Koblitz-409-a0
Koblitz-571-a0
oddext-3-2
oddext-5-2
0.0
0.2
0.4
0.6
0.8
1.0
Recorded power/GCD-shortcut time ratio (descriptive only)


Actual GF(2^83) fixed-parameter support checks
Both criteria rejected all 16; not search or root counts
Quotient-ring power
Squarefree shortcut
Recorded seconds per 16 supplied quintics
0.22
0.20
0.18
0.16
0.14
1
2
3
Internal round